

Gender Differences in Political News Coverage

This analysis explores how media portrays male and female politicians differently through language, topics, and emotional tone.

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Dataset Creation



Comprehensive Collection

News articles about 14 politicians gathered from LexisNexis archives, 150 news each.



Thorough Preprocessing

Text cleaned, politicians' names removed as custom stopwords.



Gender Balance

Equal representation of prominent female and male political figures.

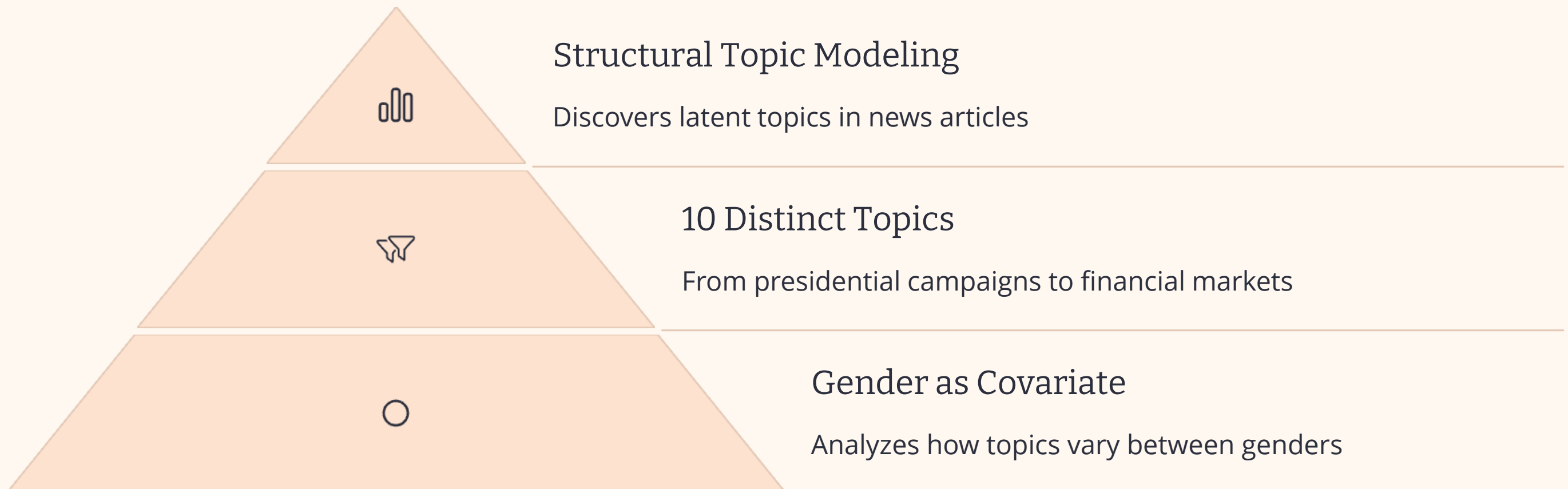
👩 Female Politicians:

- Kamala Harris
- Nancy Pelosi
- Amy Klobuchar
- Nikki Haley
- Elizabeth Warren
- Gretchen Whitmer
- Elise Stefanik

👨 Male Politicians:

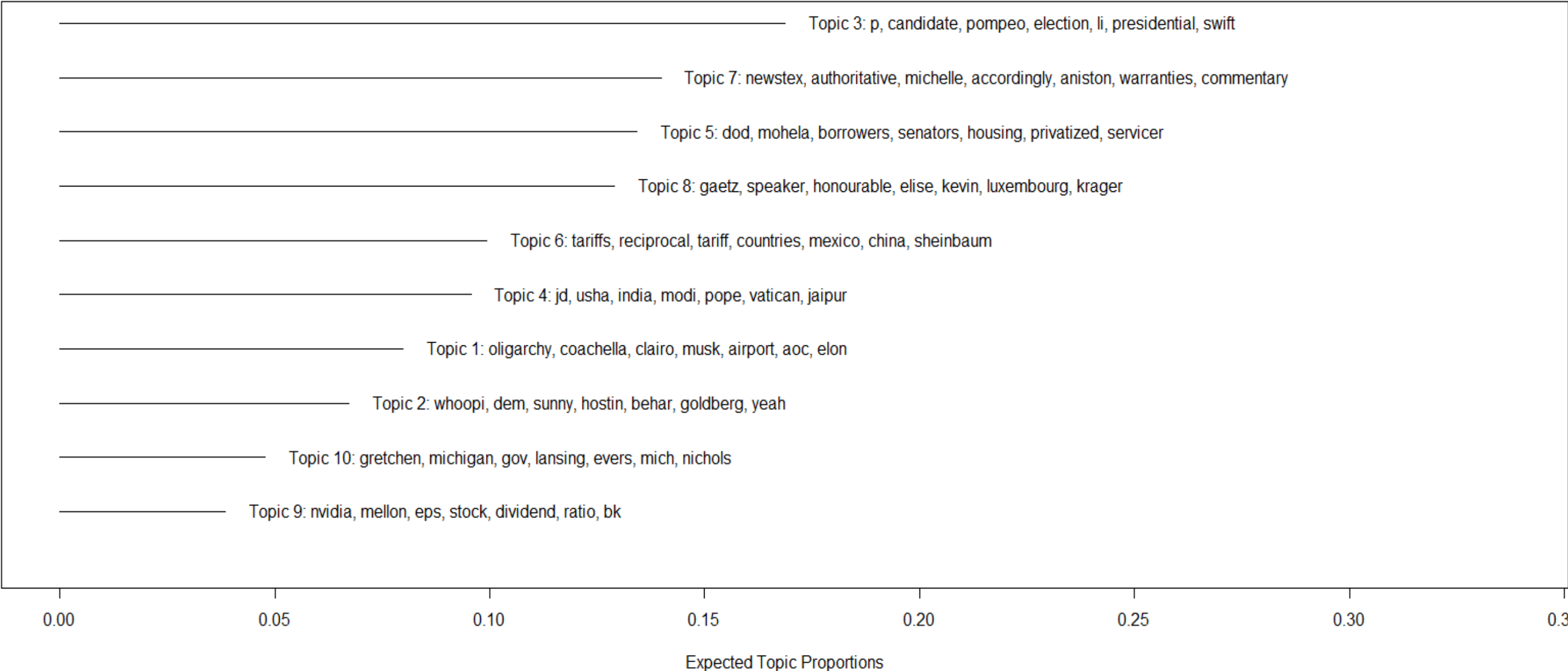
- Joe Biden
- Donald Trump
- Bernie Sanders
- Barack Obama
- Ron DeSantis
- Kevin McCarthy
- JD Vance

Topic Modeling Approach

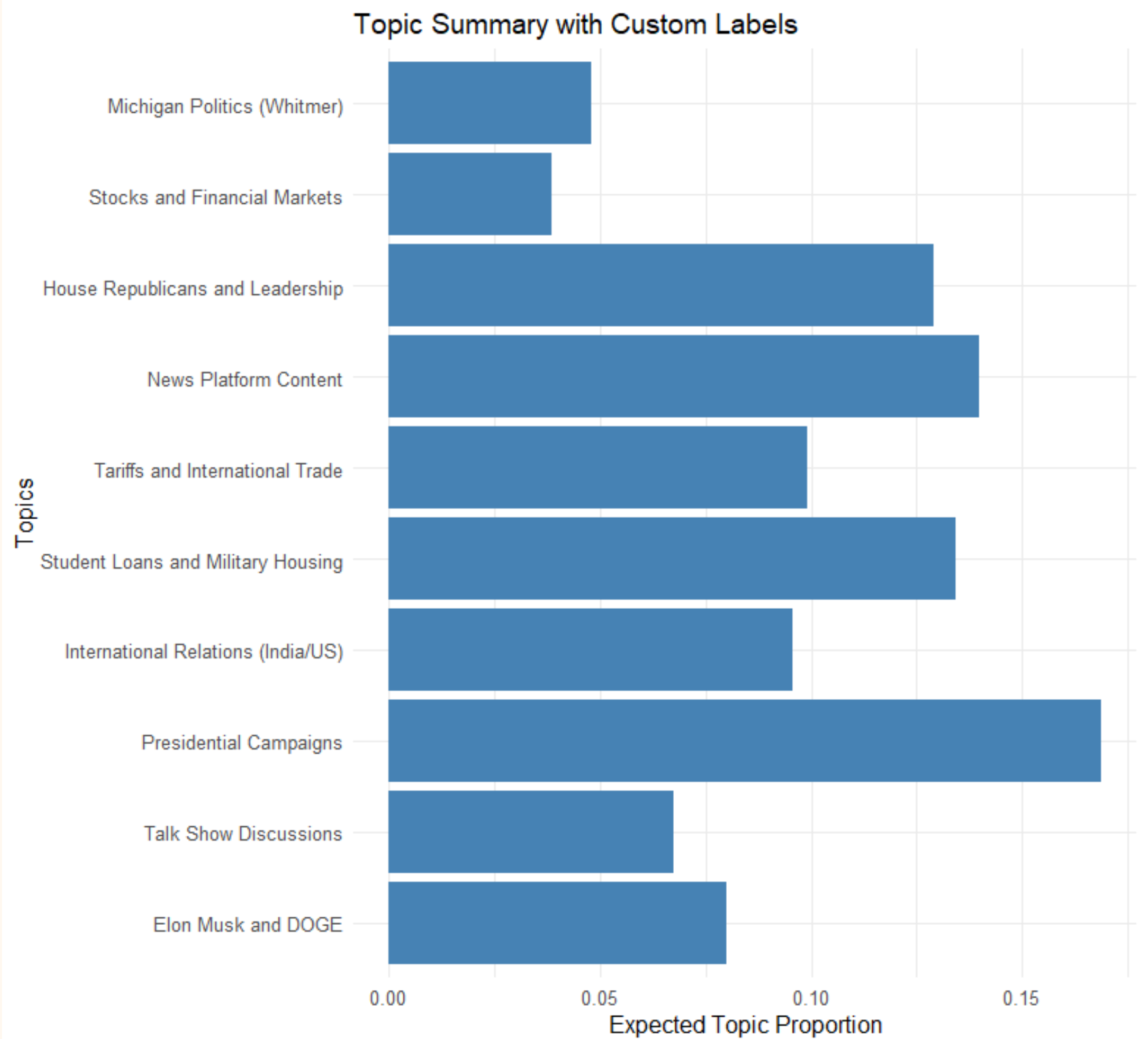


Discovered News Topics

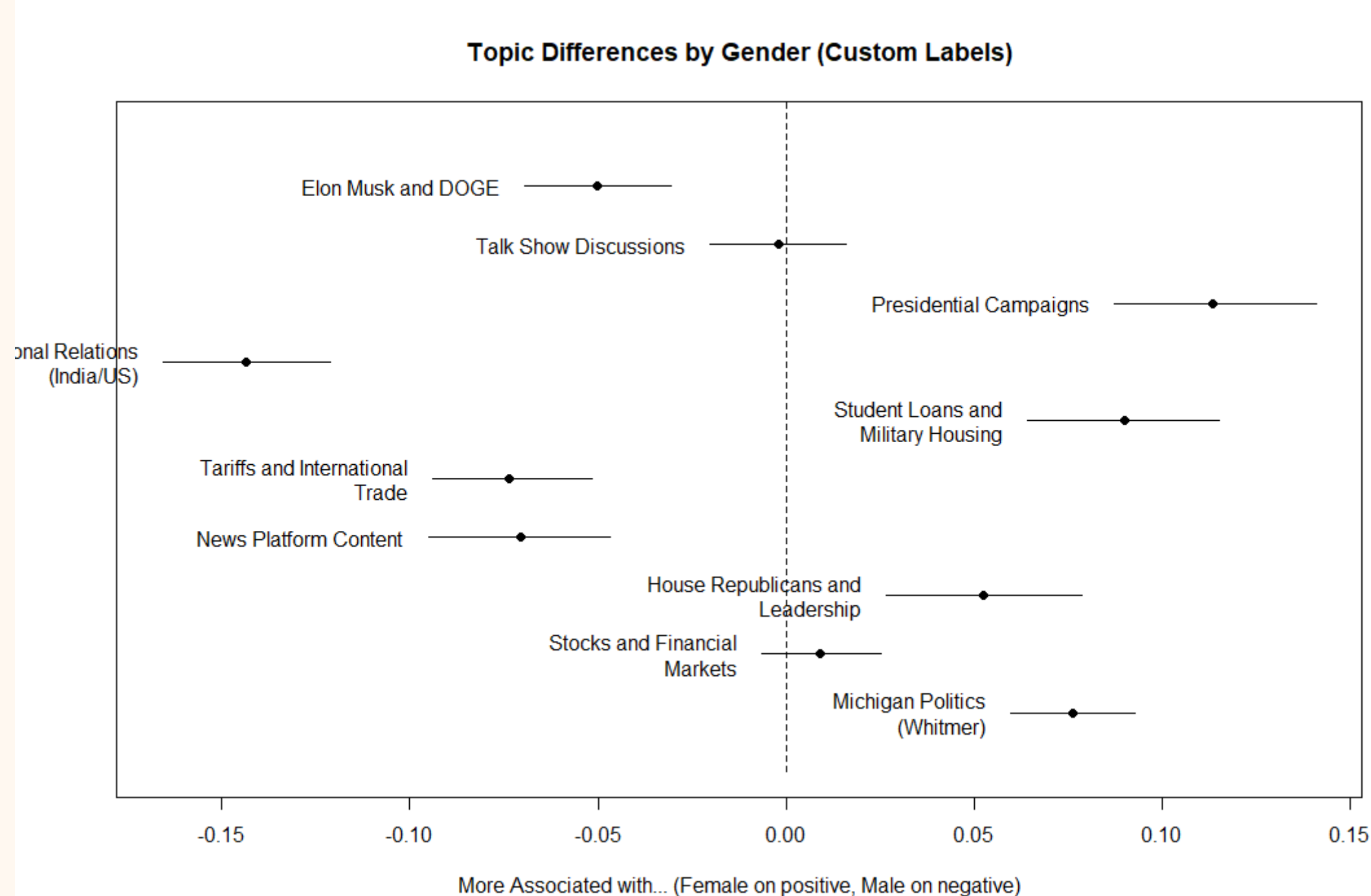
Topic Summary



Discovered News Topics



Gender Differences in Topics



Emotion Analysis

NRC Emotion Lexicon

Used established dictionary categorizing words into basic emotions.

Focus on Negative Emotions

Analyzed anger and disgust words to detect hostile coverage.

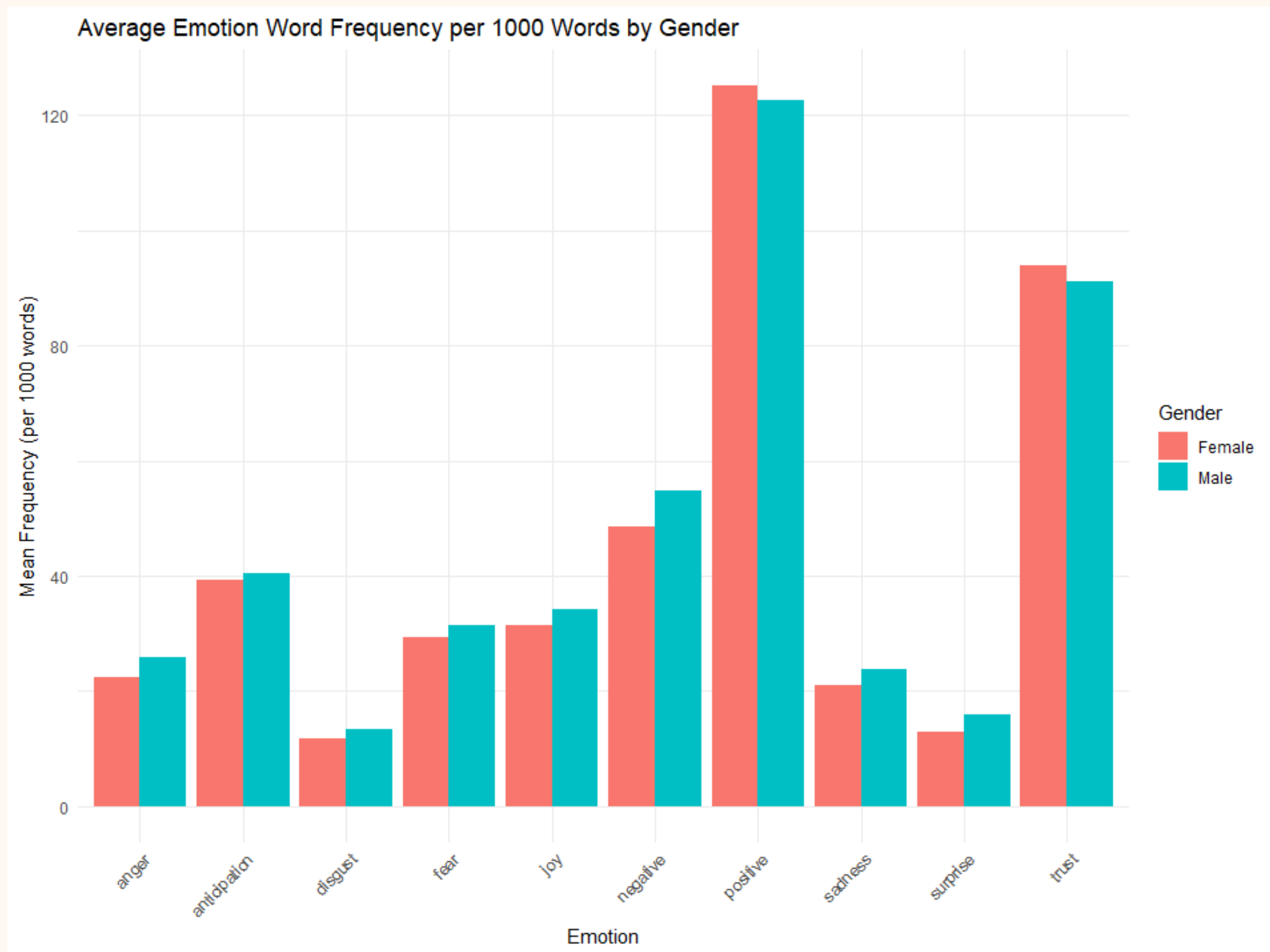
Normalized Measurement

Counted emotion words per 1,000 words for fair comparison.

Statistical Testing

Applied t-tests to assess significance of differences.

Emotion	Mean (Female)	Mean (Male)	Mean Difference	95% Confidence Interval	p-value	Significance
Anger	22.43	25.79	-3.36	[-4.94, -1.79]	2.97e-05	*** (p < 0.001)
Disgust	11.73	13.47	-1.74	[-2.75, -0.73]	0.00076	*** (p < 0.001)
Fear	29.26	31.48	-2.22	[-4.20, -0.24]	0.028	* (p < 0.05)
Sadness	20.97	23.85	-2.88	[-4.38, -1.38]	0.00018	*** (p < 0.001)





Word Embedding Analysis

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Train Word2Vec

Separate models for male and female politician articles



Create Embeddings

Words represented as vectors in semantic space



Compare Meanings

Measure similarity between same words across models



Identify Shifts

Discover words with different contextual meanings



Semantic Shift Examples

"Period"

Male: "Critical period for fiscal negotiations"

Female: "She's emotional — it's that time of the period"

"Navy"

Male: Sports or personal military experience

Female: Defense policy and oversight roles

"Woman"

Male: Relational context ("married to a strong woman")

Female: Achievement markers ("first woman to lead")

Word	Summary	Example (Male)	Example (Female)
navy	Male: army-navy games and military; Female: navy roles in political discussions and veterans.	army navy game on saturday army and navy met at northwest stadium	department of defense and the u s navy pelosi may believe amazon founder jeff bezos ...
period	Male: historical events; Female: recovery after childbirth.	for a period of rapid change in the energy industry	the period of recovery following the birth
woman	Male: 'first woman' achievements; Female: empowerment and identity.	the first woman to lead the defense department	as a woman of color she brings a new perspective to the senate
age	Male: election eligibility; Female: healthcare and maternal age.	he will reach the minimum age to qualify for the election	maternal age is a significant factor in childbirth outcomes
track	Male: campaign tracking; Female: professional achievements.	tracking the progress of the presidential campaign through each state	her track record in healthcare reform is impressive

Word	Summary	Example (Male)	Example (Female)
tension	Male: diplomatic conflict; Female: rights and healthcare tension.	tension between the us and china over trade policies escalates	tension grew over education funding cuts impacting children
jurisprudence	Male: general legal background; Female: reproductive rights jurisprudence.	jurisprudence shaped his approach to criminal justice reform	her focus on reproductive rights aligns with constitutional jurisprudence
round	Male: election rounds; Female: legislative review rounds.	he advanced to the next round in the senate elections	this bill passed the first round of legislative review
usa	Male: USA in global diplomacy; Female: USA in domestic rights discussions.	the usa imposed new tariffs during diplomatic talks	the usa must continue its commitment to civil rights
allowed	Male: military permissions; Female: civil rights and healthcare access.	soldiers were allowed to defend the border with force if necessary	women were finally allowed access to critical healthcare services



Key Findings



Implicit Bias

Even neutral words carry different meanings based on politician gender.



Topic Differences

Female politicians: more personal coverage. Male politicians: more policy focus.



Emotional Framing

Preliminary evidence suggests different emotional tones in coverage.

Thank You!!